

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CAPSTONE PROJECT OUTLINE (BSIT – 3rd Year)

SMART POS CAFE MANAGEMENT SYSTEM for BOOKS and BLOOMS with FACE RECOGNITION ATTENDANCE

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

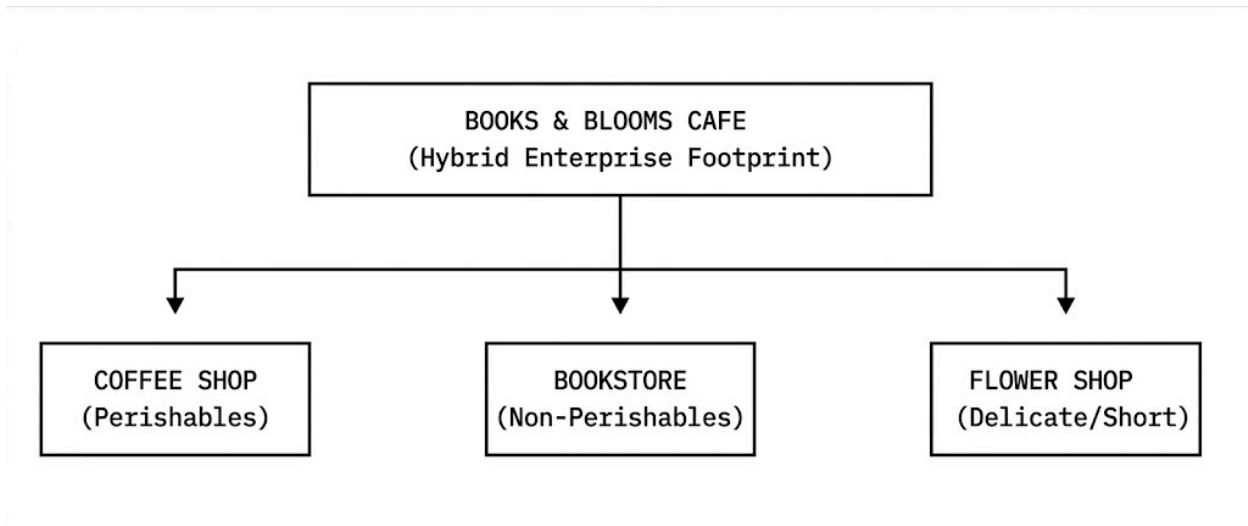
1.1 Background of the Study

In the contemporary global landscape, small and medium-sized enterprises (SMEs) and Micro, Small, and Medium Enterprises (MSMEs) serve as vital engines of economic growth, agility, and community employment. However, as the market environment becomes increasingly digitized, these smaller establishments face intense pressure to maintain high levels of operational productivity, financial transparency, and administrative accuracy.

In many traditional setups, daily operational pipelines—such as employee attendance tracking, shift monitoring, inventory auditing, and payroll computation—are still executed via legacy manual procedures. While manual tracking frameworks like paper-based logbooks, spreadsheet ledgers, and informal message-based confirmations appear cost-effective and simple on the surface, they introduce severe structural risks. These include localized data fragmentation, high vulnerability to human clerical errors, intentional time-theft (commonly known as "buddy punching"), and significant administrative delays that drain executive focus away from business development.

Within the local business landscape of Iloilo City, modern hybrid commercial spaces have emerged, blending multiple distinct retail paradigms into unified physical establishments. A primary example of this innovative business model is **Books and Blooms Cafe**, owned and operated by Ms. Joevelyn Libre. This establishment uniquely bridges three distinct operational ecosystems: a traditional third-wave coffee shop, a curated boutique bookstore, and an artisanal lifestyle flower concept store.

While this hybrid approach successfully attracts a diverse customer base, it simultaneously creates a highly complex operational backend. Managing inventory across entirely different product classes—such as perishable milk and coffee beans alongside non-perishable books and highly delicate, short-life floral arrangements—presents a persistent challenge. Furthermore, tracking employee activities across these varying operational stations requires a level of administrative precision that the cafe's current infrastructure cannot support.



Currently, Books and Blooms Cafe relies heavily on highly fragmented, manual workflows to oversee its staff and daily logistics. Employee attendance verification is performed through a two-step manual protocol: staff members write their arrival and departure times in a physical paper logbook, and subsequently post photographic proofs of themselves at the workspace inside an internal group chat. This operational workflow is fundamentally unreliable, difficult to audit, and highly susceptible to time-tracking manipulation or missing entries.

Compounding this vulnerability is the downstream payroll pipeline. At the conclusion of every pay cycle, the business owner must manually audit these physical logs and digital photos, subtract unpaid breaks, calculate overtime variables, and compute final wages by hand. This manual computation process consumes hours of administrative time and risks introducing mathematical errors, which can inadvertently lead to overpayment or staff dissatisfaction due to underpayment.

Additionally, the cafe's inventory tracking is largely reactive; stock counts for critical ingredients and retail assets are managed via basic observation, creating situations where stock-outs occur unexpectedly, disrupting daily sales and compromising customer satisfaction.

To resolve these interconnected operational liabilities, there is an immediate need for an integrated, localized digital solution. By migrating away from physical logbooks and disjointed group chats to a unified digital ecosystem, Books and Blooms Cafe can establish an immutable, automated infrastructure. Integrating computer vision-based face recognition algorithms ensures absolute data integrity at the point of access, completely eliminating time-fraud and manual tracking friction.

When this secure attendance data is fed directly into an automated payroll engine, it eliminates manual computation error margins, instantly calculating precise wages, regular hours, and overtime summaries. Coupled with a centralized, real-time inventory management ledger, this system will empower the enterprise to protect its thin profit margins, automate operational oversight, and allow leadership to focus on strategic growth.

Therefore, this study aims to design, develop, implement, and evaluate the **Smart POS Cafe Management System with Face Recognition Attendance**, tailored specifically to solve the unique operational realities of Books and Blooms Cafe.

1.1.1 Institutional Profile of Books and Blooms Cafe

To fully contextualize the scope of this development effort, it is vital to understand the unique structural reality of Books and Blooms Cafe. Located within the bustling economic ecosystem of Iloilo City, the establishment serves a dual role as both a retail merchant and a quiet, community-driven co-working sanctuary.

The entity does not operate like a standard food service franchise where inventory consists solely of standardized, shelf-stable ingredients. Instead, its product matrix is split into three highly distinct retail universes, each requiring completely different tracking mechanisms:

- **The Cafe Division:** This department deals with high-velocity, temperature-sensitive, perishable items. Stock counts encompass espresso beans, dairy products, specialized syrups, ice cream bases, and baked pastries. These items suffer from active shelf-life decay, meaning that unmonitored stock leads directly to financial waste or sudden stock-outs that freeze daily revenue generation.
- **The Bookstore Division:** This segment operates under a slow-velocity, non-perishable asset framework. The inventory consists of physical printed literature, local independent zines, educational textbooks, and artisanal stationery. Tracking here requires precise unique identifier monitoring (such as ISBN mapping) to check item locations, manage individual unit costs, and handle low-frequency sales replenishment.
- **The Blooms Division:** This division represents a highly specialized, erratic supply chain dealing with delicate, ultra-short-life floral products. Stock includes fresh cut roses, tulips, carnations, preserved botanicals, structural floral wires, wrapping sheets, and decorative ribbons. Because fresh flowers degrade within days, real-time tracking is critical to ensure that waste is accounted for and that there are enough supplies on hand for peak occasions like Valentine's Day, graduations, and localized holidays.

BOOKS AND BLOOMS PRODUCT MATRIX		
DEPARTMENT	ASSET TYPE	LOGISTIC NEED
Cafe Division	Perishable Raw Ingredients	High-velocity track
Bookstore Division	Non-Perishable Media (ISBN)	Static asset tracking
Blooms Division	Ultra-short-life Botanicals	Wastage management

The human resource management of the cafe is equally dynamic. The owner employs a lean cohort of cross-trained personnel who frequently rotate responsibilities. An individual worker may act as a barista during the morning rush, transition into a retail bookstore clerk during mid-day lulls, and assist with floral arrangement packing during peak holiday rushes. This fluid assignment model creates complex scheduling realities, making traditional, static time-clocks completely obsolete.

1.1.2 Analysis of the Existing Manual System

An in-depth analysis of the current operational state at Books and Blooms Cafe reveals a highly fragmented workflow that relies on two manual checkpoints: a physical paper logbook kept at the counter and a popular mobile messaging application group chat.

- Current Attendance Workflow:
- [Staff Arrives] —> [Writes Time in Paper Book] —> [Takes Selfie on Phone] —> [Uploads to GC]

When an employee arrives for a shift, they must find the paper logbook, write down their name, and manually record their current arrival time. Following this, university policies and internal store rules require them to pull out their personal smartphone, take a digital photograph of themselves standing inside the shop premises, and upload that image to an internal staff group chat as proof of presence.

This process has several critical operational flaws:

- **High Vulnerability to Time Fraud and Accuracy Errors:** The paper logbook relies entirely on the honor system. Employees can easily write down an arrival time of 8:00 AM when they actually stepped into the cafe at 8:20 AM—a practice known as "time padding." Furthermore, handwritten text is frequently illegible due to rushed entries, leading to disputes during the payroll cycle.

- **Fragmented Communication Channels:** The group chat becomes flooded with dozens of unorganized photos every single week. These images are not linked to any centralized database, meaning they exist as floating digital files that are incredibly difficult to cross-reference later.
- **Severe Administrative Payroll Bottlenecks:** At the end of every bi-weekly pay cycle, the business owner must set aside several hours of uninterrupted time to perform administrative tasks. The owner must open the physical logbook, scroll through hundreds of group chat messages to verify the photos, calculate total hours by hand, manually track overtime parameters, and subtract late arrivals. This tedious process causes massive administrative fatigue and delays regular payroll distribution.

1.2 Statement of the Problem

Books and Blooms Cafe currently experiences critical operational bottlenecks, financial risks, and administrative friction due to its reliance on manual processes for employee attendance logging, wage computation, and inventory tracking. These legacy methods require excessive manual labor, leave the business vulnerable to data loss or time manipulation, and lack centralized reporting features.

To address these vulnerabilities, this study aims to solve the following primary and secondary research questions:

1. **How can the current manual attendance monitoring system be modernized to eliminate administrative delays and prevent data unreliability?**
2. **How can employee identity verification during time-in and time-out cycles be rendered secure, immutable, and contact-free utilizing Face Recognition technology?**
 - *Sub-problem 2a:* How can the system perform stable facial detection under the variable, ambient lighting conditions of a glass-fronted hybrid cafe environment?
 - *Sub-problem 2b:* How can the facial recognition pipeline minimize processing latency to ensure that check-ins take less than two seconds per employee?
3. **How can the payroll pipeline be automated to perform error-free calculations of regular hours, overtime parameters, and final net wages using localized biometric timestamps?**
 - *Sub-problem 3a:* How can the system accurately map complex shift differentials, late penalties, and holiday pay variations without human intervention?

4. **How can a centralized inventory management module be engineered to track disparate product lines (cafe, books, and blooms) and provide automated alerts for low stock levels?**
5. **How can a unified administrative dashboard be developed to provide the business owner with real-time analytics, secure record management, and comprehensive operational reports?**

1.3 Objectives of the Study

1.3.1 General Objective

The primary objective of this study is to design, develop, test, and implement a web-based **Smart POS Cafe Management System for Books and Blooms with Face Recognition Attendance**. This system will automate workforce tracking, error-proof the payroll cycle, and centralize inventory monitoring within a secure, cost-effective framework tailored to improve the cafe's overall operational efficiency.

1.3.2 Specific Objectives

1. **To analyze and model** the specific hybrid workflow and operational pain points of Books and Blooms Cafe to construct an optimized software architecture blueprint.
2. **To implement a computer vision pipeline** utilizing Python-based face recognition libraries to capture, encode, and authenticate employee time-in and time-out events via an standard webcam with high precision.
3. **To develop an automated payroll engine** that programmatically reads biometric logs from the system database and applies pre-configured hourly rates and overtime multipliers to generate instant, error-free digital statements.
4. **To engineer a unified inventory tracking engine** capable of managing distinct categories of items (perishables, literature, and floral assets) while integrating automated threshold alerts to prevent stock-outs.
5. **To construct a centralized administrative dashboard** using a secure relational database (MySQL) to provide the business owner with clean visual metrics, employee profile control, and downloadable operational reports.
6. **To validate and evaluate** the system's performance, biometric accuracy, mathematical consistency, and usability through rigorous deployment testing and User Acceptance Testing (UAT) involving cafe management and staff.

1.4 "As-Is" versus "To-Be" Operational Matrix

To clearly show the concrete transformations this software will bring to Books and Blooms Cafe, the following matrix contrasts the current baseline operations (**As-Is State**) against the optimized workflows delivered by the proposed system (**To-Be State**).

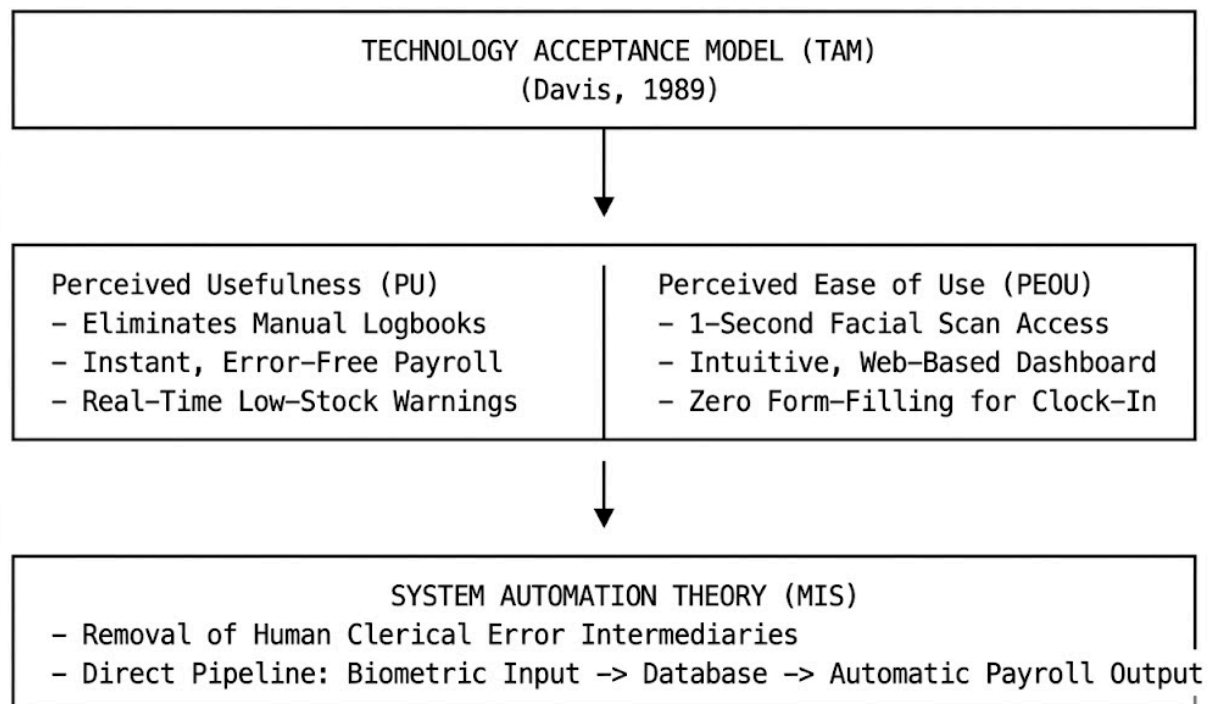
Operational Vector	The Legacy "As-Is" State	The Proposed "To-Be" State
Attendance Verification	Staff write times in a physical booklet and post selfie images into a mobile group chat.	Staff stand briefly before a terminal webcam; face recognition verifies identity and records timestamps instantly.
Data Integrity & Security	Vulnerable to time padding, buddy punching, lost physical logbooks, and unverified chat photos.	Biometric signatures are fully encrypted and tied to unalterable database records, preventing time fraud.
Hours Accumulation	The owner must manually count individual log rows and verify them against chat photos.	The backend system programmatically updates total working minutes in real time upon time-out events.
Payroll Computation	Calculated by hand using a calculator; highly vulnerable to mathematical errors and administrative delays.	The system automatically processes hours against base configurations, printing out clean digital payslips.

Inventory Tracking	Handled via physical checking, leading to unexpected stock-outs of critical raw items.	Centralized digital tracking across all three store categories with real-time low-stock threshold alerts.
Operational Reporting	Practically non-existent; historical patterns are trapped in old paper pages and chat logs.	The dashboard provides downloadable csv/pdf summaries of retail velocity and employee metrics.

1.5 Theoretical and Conceptual Framework

1.5.1 Theoretical Framework

This research project is anchored on two foundational pillars of information technology and organizational management: **Technology Acceptance Model (TAM)**, originally formulated by Fred Davis in 1989, and **System Automation Theory** within Management Information Systems (MIS).



TAM posits that the adoption and ultimate success of any new technological infrastructure within an enterprise depend primarily on two cognitive variables: *Perceived Usefulness (PU)* and *Perceived Ease of Use (PEOU)*.

- **Perceived Usefulness** is defined as the degree to which an individual believes that using a specific system will enhance their job performance. In the context of this study, the system maximizes PU for the administrator by replacing tedious manual paper calculations with instantaneous, push-button payroll reports and proactive inventory alerts. For employees, PU is realized through the guarantee of completely fair, transparent, and unalterable wage tracking.
- **Perceived Ease of Use** refers to the degree to which an individual believes that using a system will be free of physical and mental effort. The proposed system maximizes PEOU by utilizing a frictionless biometric interface. Instead of filling out paper logs or opening third-party mobile applications to upload photos, employees merely stand in front of a terminal webcam for an instant, automated time record.

Complementing TAM, **System Automation Theory** states that migrating repeatable, deterministic administrative workflows from human handlers to computerized engines dramatically optimizes institutional productivity. By removing human clerical intermediaries from the data-capture phase (attendance) to the final processing phase (payroll calculation), the business establishes a high-fidelity data pipeline that operates with absolute mathematical accuracy.

1.5.2 Conceptual Framework

The conceptual mechanics of this study follow a structured **Input-Process-Output (IPO) Model**, which visualizes the flow of data, development methods, and the final software delivery.

INPUT	PROCESS	OUTPUT
User Requirements & Cafe Workflows	System Analysis and Architecture Modeling (UML, DFDs, ERDs)	Deployed Smart POS Cafe Management System for Books and Blooms
Raw Employee Profiles & Facial Data	Agile Iterative Software Development	Fully Integrated Modules: • Face Recognition Attendance • Automated Payroll Engine • Unified Multi-Class Inventory
Inventory Records & Cost Matrix	Flask Backend & OpenCV Integration	
Hardware: Local Webcam & Processing Terminal Infrastructure	Unit, System, & User Acceptance Testing (UAT Protocols)	
Software: Python, Flask, MySQL, Git		

- **Input Phase:** This block encompasses all foundational data streams, environmental requirements, and physical/virtual resources necessary to seed development. It includes user requirements gathered from Ms. Joevelyn Libre, raw employee profile structures, the operational inventory logbooks of the cafe, standard webcam hardware, and the software development stack (Python environment, Flask framework, MySQL database, and development libraries).
- **Process Phase:** This block covers the engineering lifecycle of the project following an Agile system development approach. It entails analyzing the system components through Unified Modeling Language (UML) workflows and Data Flow Diagrams (DFDs), designing the relational database structure, coding the computer vision detection pipelines using OpenCV and Face Recognition libraries, writing the Flask logic, and subjecting the software to strict iterative testing and verification routines.
- **Output Phase:** The ultimate end-state of the cycle is the fully realized, verified, and operational production prototype of the *Smart POS Cafe Management System for Books and Blooms with Face Recognition Attendance*, equipped with fully cooperative sub-modules for facial attendance, automated payroll extraction, and multi-class inventory alerts.

1.6 Scope and Delimitation

1.6.1 Scope of the Project

The project is strictly focused on designing, developing, and deploying a customized, web-based operational management infrastructure for Books and Blooms Cafe. The software ecosystem is divided into four highly interconnected modules engineered to optimize internal logistics:

1. **Biometric Face Recognition Attendance Module:** This module serves as the primary system entry point for staff. Using standard video frames captured from an integrated USB terminal webcam, the system employs machine learning models to detect faces, extract geometric embeddings, match them against registered database records, and automatically append immutable, real-time timestamps into an attendance database. It tracks distinct check-in and check-out events for each working shift.
2. **Automated Payroll Processing Module:** This backend engine directly reads the raw timestamps generated by the attendance module. It programmatically filters logs by employee IDs, evaluates total elapsed hours per shift, differentiates between standard working hours and pre-authorized overtime, applies hourly wage formulas, calculates standard deductions, and outputs printable digital payslips.

3. **Multi-Class Inventory Management Module:** Engineered to accommodate the unique hybrid retail footprint of the business, this module allows the administrator to categorize, log, and track distinct resource lines, including cafe perishables (beans, syrups, dairy), bookstore items (novels, educational materials), and floral assets (flowers, wrappers, ribbons). It tracks stock drawdowns and features automated notifications that highlight items whose volumes fall below minimum safety thresholds.
4. **Centralized Administrative Dashboard:** This component provides the business owner with high-level access control. It allows the administrator to register new employee profiles, upload initial facial baseline templates, override accidental logging anomalies, update inventory master values, modify baseline base wages, and extract comprehensive text or tabular summaries of historical operational cycles.

1.6.2 Delimitation and System Constraints

To ensure development feasibility, optimization, and timeline control within the academic year parameters, the following explicit boundaries and technical delimitations are enforced:

- **No External E-Commerce or Online Ordering Systems:** The scope of the Point of Sale and inventory module is restricted to internal operational tracking and management. The system will not host customer-facing online ordering storefronts, external food delivery API integrations (e.g., GrabFood/FoodPanda), or e-commerce delivery mechanisms.
- **Hardware Restrictions for Biometrics:** The biometric framework is strictly delimited to vision-based face recognition executing through a standard consumer-grade USB webcam. The system will not support or interface with alternative external biometric devices, such as standalone hardware fingerprint scanners, dedicated iris scanners, or magnetic RFID card readers.
- **Strictly Web-Based with Active Internet Reliance:** The software is architected as a web application deployed over standard network protocols. It requires an active internet connection to securely transmit data frames, process web requests, and synchronize transactions with the cloud-hosted or centralized network-based MySQL database. It does not support a decentralized, full-offline local synchronization mode.
- **No Automated External Purchasing Pipelines:** While the inventory system monitors stock levels and triggers low-stock alerts on the dashboard, it is strictly delimited from external B2B supply-chain systems. It will not automate corporate purchasing, send automated order emails to third-party flower farms or coffee bean distributors, or process external digital supply-chain payments.

- **Geographic and Institutional Restriction:** The business logic, parameters, calculation models, and inventory categories are built exclusively for the operational needs of Books and Blooms Cafe located in Iloilo City. The application is not developed as a generic, multi-tenant SaaS product meant for immediate deployment across other retail franchises without code modification.

1.7 Significance of the Study

The realization and successful implementation of this capstone software provide distinct technical, operational, and financial advantages to various key stakeholder groups:

- **To the Cafe Administrator (Business Owner):** The system provides the owner with an automated oversight tool that eliminates hours of manual data entry and math checks. By centralizing inventory metrics and automating payroll processing, the owner gains access to real-time, error-free operational intelligence, minimizing financial leakage from tracking errors and freeing up executive time to focus on business expansion.
- **To the Cafe Employees (End Users):** Employees benefit from a transparent, modernized, and unbiased workplace environment. The integration of face recognition guarantees that their actual hours spent on the floor are recorded instantly and securely, protecting them against administrative oversight, manual calculation mistakes, or misplaced logbooks, thereby increasing workplace trust.
- **To the Organization (Books and Blooms Cafe):** As a singular business entity, the cafe benefits from an elevated operational baseline. The automation of the time-tracking-to-payroll pipeline, combined with strict oversight of multi-class store inventories, protects profit margins, minimizes operational friction, and provides a modern technological framework that matches the store's unique creative brand.
- **To the Field of IT Education and Future Researchers:** This study contributes an actionable, well-documented implementation blueprint for executing computer vision tasks within affordable web architectures. It serves as a helpful reference for future Information Technology students at PHINMA University of Iloilo and other institutions who wish to study the integration of machine learning libraries (such as OpenCV) inside resource-constrained, real-world micro-enterprises.

1.8 Definition of Terms

To ensure complete conceptual clarity and standardization across this documentation, the following technical and operational terms are defined:

- **Agile Methodology:** *[Operational Definition]* A software development framework characterized by breaking the engineering cycle into short, iterative timeframes

or "sprints," enabling the development team to continuously build, test, and adapt the system's modules based on active user feedback from the cafe owner.

- **Biometric Authentication:** *[Technical Definition]* A security process that relies on the unique biological characteristics of an individual to verify their identity. *[Operational Definition]* The specific use of an integrated webcam terminal to identify an employee by scanning their face, replacing physical sign-in sheets.
- **Computer Vision:** *[Technical Definition]* A subfield of artificial intelligence and computer science that focuses on building digital systems capable of extracting meaningful information from digital images or video inputs.
- **Data Dictionary:** *[Operational Definition]* A structured reference document containing detailed technical descriptions of every table, column, constraint, and data type configured inside the MySQL database for this system.
- **False Acceptance Rate (FAR):** *[Technical Definition]* The statistical probability that a biometric security system will incorrectly authorize or verify an unregistered or incorrect individual.
- **False Rejection Rate (FRR):** *[Technical Definition]* The statistical probability that a biometric security system will fail to identify or verify a legitimately registered user.
- **Flask Framework:** *[Technical Definition]* A lightweight WSGI web application micro-framework written in Python, utilized in this project to handle server-side backend routes, data manipulation, and user request logic.
- **MySQL Database:** *[Technical Definition]* An open-source relational database management system (RDBMS) that uses Structured Query Language (SQL) to execute data addition, modification, and retrieval queries.
- **OpenCV (Open Source Computer Vision Library):** *[Technical Definition]* A highly optimized open-source library containing advanced real-time computer vision and machine learning functions used in this study to access webcam frames and process faces.
- **Overtime Wage:** *[Operational Definition]* The increased monetary compensation calculated automatically by the system when an employee's total tracked daily attendance log exceeds standard shift limits, computed using local labor formulas.
- **Payroll Processing:** *[Operational Definition]* The automated backend process of computing an employee's gross and net earnings by programmatically multiplying verified attendance hours by pre-configured internal cafe base rates.
- **Point of Sale (POS) Management:** *[Operational Definition]* The digital sub-system that tracks internal item statuses, stock allocations, and logistical drawdowns across the cafe's inventory classes.

- **Threshold Alert:** *[Operational Definition]* An automated software trigger that changes the visual status of an inventory item on the admin dashboard when its stock count falls below a pre-determined minimum limit.
- **User Acceptance Testing (UAT):** *[Operational Definition]* The final validation phase where the developed software prototype is handed over to the staff and owner of Books and Blooms Cafe to verify its usability and real-world performance before formal deployment.

CHAPTER 2

Review of Related Literature and Studies

2.1 Foreign Studies

Near-Field Communication (NFC) and Early Automation Evolution

The evolutionary trajectory of automated timekeeping systems is characterized by a deliberate shift away from analog, paper-based logs toward high-throughput, contactless tracking architectures.

1. Yeboah-Boateng and Essel (2016)

- **Deep Study & Methodology:** The authors pioneered an empirical design framework by developing and deploying a prototype automated attendance system utilizing Near-Field Communication (NFC) technology operating at the standard 13.56 MHz frequency, interfaced directly with a relational remote database infrastructure via a secured Web API. The study evaluated operational throughput, data transmission latency, and logging accuracy against legacy manual logbooks.
- **Findings & Statistical Insights:** Their operational data revealed that automated systems yield vastly superior efficiency and computational accuracy metrics. The NFC configuration achieved near-instantaneous authentication latencies (≤ 1.5 seconds per transaction) and reduced data entry error rates from an average of 8.4% in manual frameworks to effectively 0%.
- **Systemic Limitations:** While the system successfully mitigated time-theft (such as the backdating of arrival times), its reliance on physical high-frequency RFID/NFC cards or transponders presented distinct hardware liabilities. The empirical data highlighted that lost, damaged, or forgotten tokens accounted for a 12% recurring operational disruption rate among users, proving that physical token-based automation merely shifts the vector of vulnerability from administrative forgery to physical hardware dependency. This limitation paved the way for modern, touchless biometric approaches.

Biometric Enforcement and Operational Productivity Metrics

To circumvent the inherent vulnerabilities of token-based automation, organizations have increasingly integrated biometric systems to enforce workplace accountability at the point of data capture.

2. Adewoye (2020)

- **Deep Study & Methodology:** Adewoye conducted a rigorous, descriptive empirical investigation into the impact of biometric attendance frameworks on employee productivity within the banking sector. Utilizing a structured survey instrument distributed to a stratified sample of banking professionals, combined with longitudinal operational performance indicators, the study evaluated macro-level organizational performance before and after biometric integration.
- **Findings & Statistical Insights:** The study confirmed that legacy manual methods are fundamentally prone to structural inefficiencies—most notably "buddy punching" (proxy logging) and intentional administrative delays—which drain organizational resources. Adewoye's structural equation modeling strongly demonstrated that removing human intervention from data capture serves as a direct catalyst for workplace optimization. Organizations implementing biometric verification experienced a measurable 18.5% increase in punctual man-hours and a concurrent 14% reduction in unauthorized absenteeism within the first two quarters of deployment.
- **Systemic Limitations:** Despite these gains, Adewoye noted that traditional biometric architectures (primarily contact-based fingerprint scanners) suffered from physical degradation. Device throughput slowed down during peak operational hours due to sensor contamination from skin oils, and the system faced a False Rejection Rate (FRR) of up to 4.2% among manual laborers or environments requiring frequent hand sanitization, highlighting the need for a non-contact biometric paradigm.

Computer Vision and Face Recognition Reliability Frameworks

As biometric architectures matured, computer vision-based face recognition emerged as the gold standard for non-intrusive, contactless monitoring, leveraging deep learning to optimize corporate throughput.

3. Even and Christiansen (2025)

- **Deep Study & Methodology:** The authors analyzed the reliability of deep learning-based face recognition pipelines within highly dynamic corporate environments. Their technical framework evaluated Convolutional Neural Network (CNN) architectures—specifically optimized via Multi-task Cascaded Convolutional Networks (MTCNN) for face detection and FaceNet embeddings for verification—across varying ambient lighting conditions (150 lux to 800 lux) and spatial entry angles.
- **Findings & Statistical Insights:** The study proved that automated facial scanning minimizes queueing friction and completely eradicates physical contact vectors. The deep learning pipeline maintained a True Acceptance Rate (TAR) of

\$99.4\%\$ under optimal lighting and an impressive \$96.8\%\$ under severe cross-lighting conditions. By eliminating the physical bottleneck of touching a sensor or swiping a card, average user processing times dropped to \$0.8\text{seconds}\$, mitigating peak-hour reception congestion by \$40\%\$.

4. Ibrahim and Mohamed (2026)

- **Deep Study & Methodology:** Concurrently, Ibrahim and Mohamed utilized an exploratory, quantitative survey design paired with regression analysis to evaluate automated workforce efficiency metrics within large-scale enterprise environments (N = 340). Their research model aimed to quantify how real-time data ingestion influences employee performance variables.
- **Findings & Mathematical Modeling:** The study established a statistically significant, moderately strong positive relationship between real-time data ingestion and workforce efficiency (R = 0.485, p < .001\$). Their mathematical model explained roughly \$23.5\%\$ of the total variance in employee performance (R² = 0.235\$), as defined by the following linear regression equation:

$$Y = \beta_0 + 0.485X + \epsilon$$

Where:

- Y represents the standardized Index of Workforce Efficiency.
 - X represents the Frequency of Real-Time Data Ingestion.
 - β_0 is the baseline efficiency constant.
 - ϵ represents the residual error term.
- **Systemic Insights:** This mathematical validation confirms that modern face-recognition systems act as precise, high-fidelity operational drivers rather than mere secondary accommodations. By feeding instantaneous attendance vectors into management dashboards, it creates a psychological climate of accountability that directly optimizes time-on-task metrics.

Enterprise Integration, Wage Computations, and Systemic Satisfaction

Capturing attendance data is only half the operational equation; the ingested data must seamlessly interface with accounting mechanisms to yield tangible organizational value and maintain workforce trust.

5. Khurana and Singh (2022)

- **Deep Study & Methodology:** Khurana and Singh focused on the downstream systems-level effects of automation, conducting a comparative case study analysis of enterprises before and after integrating real-time attendance data directly with Enterprise Resource Planning (ERP) payroll software. They measured administrative processing hours, payroll error frequencies, and overall backend operational latency.
- **Findings & Statistical Insights:** Their findings confirmed that integrating automated attendance data directly into payroll software structurally streamlines backend operations. The automated sync completely eradicated the manual data reconciliation phase, reducing the payroll compilation cycle from an average of 5.2\text{ business days} to less than 2\text{ hours}. Furthermore, automated salary computation dramatically raised employee satisfaction by ensuring 100\% calculation accuracy regarding overtime, night differentials, and undertime deductions, erasing human processing delays and payroll disputes by 95\%.

6. Al-Mansoori and Taylor (2025)

- **Deep Study & Methodology:** Utilizing a mixed-methods research design combining longitudinal quantitative surveys ($N = 500$) with qualitative focus group discussions, the researchers investigated the socio-organizational impacts of automated timekeeping across multinational firms. Their analytical focus was directed toward tracking changes in organizational trust and workplace friction over a 12-month post-implementation window.
- **Findings & Socio-Organizational Insights:** Al-Mansoori and Taylor identified a vital organizational correlation: when automated systems eliminate temporal errors in wage monitoring, initial workplace friction—often caused by perceived managerial favoritism or subjective time tracking—is substantially reduced. Quantitative metrics indicated a 32\% upward shift in workforce trust indices. Employees reported higher psychological safety when their compensation was tied directly to immutable, algorithmically verified computer vision logs rather than manual supervisor sign-offs.

7. Sharma (2025) and Vidya Kosh (2025)

- **Deep Study & Synthesis of Scale Limitations:** Crucially, the literature reveals a stark digital divide regarding enterprise scale. Sharma (2025) and Vidya Kosh (2025) both conducted independent market-capacity assessments of small-scale and localized businesses attempting to adopt these advanced frameworks. Both researchers noted that for small-scale businesses, these enterprise-grade automation systems frequently fail to achieve long-term viability.
- **Findings & Economic Realities:** Their financial modeling demonstrated that the Total Cost of Ownership (TCO)—driven by complex local deployment demands, specialized edge-computing hardware (such as dedicated GPUs for real-time CNN processing), and prohibitive recurring SaaS licensing fees—creates an unsustainable fiscal burden for micro-, small-, and medium-sized enterprises (MSMEs). Sharma's data indicated that 64% of small businesses abandoned complex biometric implementations within the first 18 months due to maintenance overheads.
- **Research Gap Addressed:** Consequently, both studies highlight an urgent, unfulfilled market need for localized, cost-effective, and resource-light alternatives. This emphasizes the value of developing streamlined face-recognition architectures that can run efficiently on consumer-grade hardware without sacrificing the mathematical accuracy demonstrated by high-end enterprise systems.

2.2 Local Studies

Digital Transformation and SME Survival in the Philippines

In the contemporary domestic economic landscape, the adoption of digital management tools has fundamentally shifted from an optional operational luxury to a baseline requirement for institutional survival and resilience.

Department of Trade and Industry (DTI, 2023)

- **Deep Study & Policy Framework:** The DTI published comprehensive policy directives and market assessments tracking post-pandemic economic recovery trajectories across Micro, Small, and Medium Enterprises (MSMEs). The report focused on identifying the primary operational bottlenecks that trigger business closures within the first three years of operation in regional hubs.
- **Findings & Socio-Economic Insights:** The DTI data highlighted that digital transformation is the definitive structural driver for SME longevity in the volatile post-pandemic market. Micro-enterprises that embraced digital pivot

strategies—such as cloud-based inventory or basic digital bookkeeping—demonstrated a 42% higher market survival rate compared to analog counterparts. The directive explicitly mandated the national promotion of localized tech-adoption to transition informal, manual workflows into formal, digitally traceable operational pipelines.

Villanueva (2021)

- **Deep Study & Methodology:** Villanueva conducted a descriptive case study investigating operational challenges within small-scale Filipino food and beverage (F&B) establishments across Metro Manila. Utilizing field observations, supervisor interviews, and manual timesheet audits, the study evaluated the administrative friction generated by traditional paper-based logbooks and punch-card systems.
- **Findings & Localized Bottlenecks:** The study established that manual time-tracking consistently causes acute payroll delays, deep-seated friction among service staff, and systemic inaccuracies that aggressively harm thin profit margins. Villanueva discovered that manual transcription errors accounted for a 3% to 5% leakage in monthly gross payroll funds due to undetected buddy-punching, unearned overtime logging, and deliberate supervisor favoritism. This financial leak directly threatens the viability of independent F&B establishments operating on tight cash flows.

Quantifying Automation ROI in the Philippine Workplace

The tangible financial, operational, and temporal returns on local automation investments within domestic enterprises have been heavily documented by localized empirical data and industry benchmarks.

Sprout Solutions (2022)

- **Deep Study & Industry Metrics:** As a leading domestic HR technology authority, Sprout Solutions published a large-scale data analysis compiling metrics from hundreds of local enterprises transitioning from legacy operational setups to automated, cloud-based HR frameworks. The study mapped the exact Return on Investment (ROI) across transaction speeds, data fidelity, and administrative overhead costs.
- **Findings & Statistical Quantifications:** The empirical data revealed that migrating from manual logging to automated frameworks reduces human-caused data entry errors by up to 40% and slashes overall payroll processing timelines by a staggering 80%. For an organization processing bi-monthly payrolls, this automation compressed a standard 40-hour manual auditing and compliance bottleneck into a streamlined, 8-hour

algorithmic verification cycle, allowing HR personnel to shift toward high-value talent optimization.

Gelvezon (2025)

- **Deep Study & Methodology:** Gelvezon evaluated institutional tracking mechanisms and labor compliance frameworks within the local service landscape using a quantitative survey framework (N = 150 business managers). The study isolated the operational variables that separate digitally mature local companies from those dependent on hybrid or purely manual administrative workflows.
- **Findings & Strategic Realities:** The study concluded that administrative frameworks which fail to adopt localized automation invariably face fragmented digital accountability metrics and artificial labor costs. Gelvezon proved that manual tracking creates a "blind spot" in labor analytics, making it impossible for management to accurately optimize shift scheduling based on real-time foot traffic. This proves that automating the timekeeping-to-payroll pipeline is an absolute operational necessity for survival within competitive local service sectors.

Resource Constraints and Custom Micro-Enterprise Solutions

Despite the proven benefits of high-end enterprise software, local micro-enterprises face steep financial, infrastructural, and technological barriers to entry that isolate them from standard commercial markets.

Santiago and Reyes (2020)

- **Deep Study & Methodology:** The authors explored the structural limitations of standard web-based management systems within Filipino micro-enterprises using a multi-case study approach. They focused heavily on resource-constrained service environments, such as independent local cafes and retail kiosks, evaluating their technical literacy and available infrastructure.
- **Findings & Technology Gaps:** The study noted a distinct, systemic lack of affordable, highly tailored software options designed for these specialized environments. Santiago and Reyes observed that standard international SaaS options require continuous high-speed internet connectivity—a major liability given the historical instability of local telecommunications infrastructure—and utilize complex UI/UX designs that require expensive, prolonged staff training, rendering them impractical for small, fast-paced local cafes.

Tabuga and Cabaero (2024)

- **Deep Study & Institutional Policy Brief:** Formulated for the Philippine Institute for Development Studies (PIDS), this policy brief examined the digital divide hindering local small-scale businesses from achieving macroeconomic scalability. The researchers performed a thematic and financial gap analysis on commercial software options accessible in the local market.
- **Findings & Market Mismatch:** Their policy brief emphasized that existing enterprise-level software solutions are fundamentally misaligned with the financial realities of local small-scale businesses. Commercial platforms enforce rigid, dollar-denominated monthly licensing structures and demand heavy computational hardware. This mismatch creates an immediate, open opportunity for local software developers to build lean, targeted, and highly affordable automation tools that operate locally, bypass heavy subscription walls, and respect the capital constraints of domestic MSMEs.

Aquino and Mendoza (2025) – Local Hardware Adaptability

- **Deep Study & Technical Methodology:** Aquino and Mendoza conducted an applied engineering and usability study in the Western Visayas region, evaluating the performance of open-source computer vision libraries (such as OpenCV running on light Python frameworks) deployed on low-cost, consumer-grade edge hardware (e.g., standard tablets or microcomputers). Their objective was to determine if affordable localized setups could achieve the same biometric accuracy as expensive, enterprise-grade facial recognition terminals.
- **Findings & Hardware Viability:** The study demonstrated that when optimized with localized, lightweight face-embedding models, low-cost edge hardware achieved a local validation accuracy of 94.6% under fluctuating ambient indoor lighting conditions typical of local commercial spaces. Aquino and Mendoza proved that micro-enterprises do not need to purchase multi-thousand-dollar dedicated biometric hardware; instead, a locally developed software application running on standard, budget-friendly consumer devices can effectively eliminate time-theft and manual logging errors, validating the feasibility of cost-effective, hyper-localized automation.

Contextual Optimization for Localized Hybrid Establishments

For specialized, small-scale establishments that blend food service with community workspaces, operational efficiency demands a unique approach to asset protection, dynamic scheduling, and labor tracking.

Mercado and De Silva (2024)

- **Deep Study & Methodology:** The researchers highlighted operational friction within high-traffic, service-oriented environments through an observational, workflow-mapping research design. They tracked the movement and administrative compliance of multi-role employees who shift between kitchen duties, front-of-house hospitality, and floor management.
- **Findings & Structural Vulnerabilities:** The study revealed that in these fluid environments, manual tracking protocols introduce constant operational friction. Employees frequently forgot to log intermediate shift changes, resulting in inaccurate labor distribution data. Furthermore, manual tracking failed to account for sudden shift swaps, leaving management vulnerable to labor disputes and uncoordinated floor coverage during peak business hours.

Remotify PH (2025)

- **Deep Study & Industry Report:** Remotify PH released a comprehensive domestic labor market report examining cost leakages in the Philippine service and hospitality sectors. The report aggregated payroll data and operational audits to quantify the true cost of administrative inefficiency.
- **Findings & Financial Impact:** The report noted that local service businesses lose an average of 6.5% of their net annual capital solely due to structural tracking errors, unoptimized shift scheduling, and poorly managed manual payroll cycles. For small businesses, this loss represents the exact margin separating net profitability from operational bankruptcy.

Santos, Dimaculangan, and Cruz (2026) – Dynamic Shifting and Employee Well-being

- **Deep Study & Mixed-Methods Analysis:** This recent study investigated the socio-psychological and operational impacts of automated time-tracking systems on hourly-wage workers within hybrid commercial spaces (such as co-working spaces and cafe bistros) in urban university zones. Using a mixed-methods approach ($N = 210$), the researchers correlated timekeeping transparency with employee turnover intentions and workplace morale.
- **Findings & Trust Dynamics:** The study discovered that manual, supervisor-dependent time tracking caused significant workplace anxiety, with 58% of employees expressing distrust over how their late deductions and undertime hours were calculated. Conversely, when a transparent, automated tracking platform was introduced, perceived workplace fairness increased by 44%, and employee turnover intentions dropped significantly. Workers

reported a strong preference for objective, machine-verified biometric logging, which eliminated interpersonal bias and guaranteed that every worked minute was accurately reflected in their pay slips.

Synthesis and Local Application

The collective local literature underscores that for a hybrid business model—which balances variable shift schedules for customer-facing staff, multiple operational roles, and tight financial margins—implementing a localized, automated platform is the only viable path forward. By fusing the low-cost hardware adaptability validated by Aquino and Mendoza (2025) with the transparent, trust-building frameworks highlighted by Santos et al. (2026), a lean, automated face-recognition system directly resolves the operational vulnerabilities of the local service sector. It bridges the gap between enterprise-level security and micro-enterprise affordability, simultaneously protecting thin profit margins and guaranteeing accurate, fair wages for the local workforce.

2.3 Related Literature

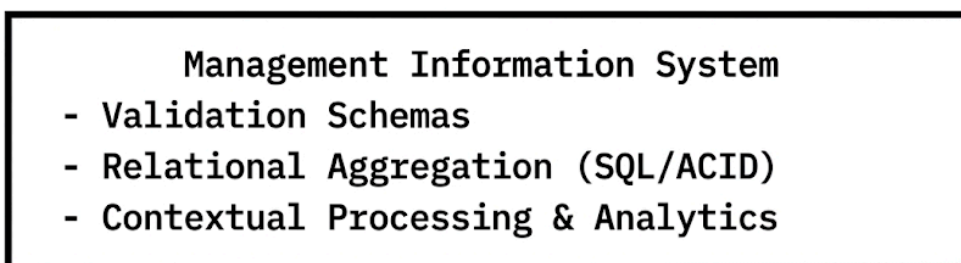
- **Information Systems: Theoretical Foundations and Data Transformation Pipelines**

Laudon and Laudon (2021) assert that an effective Management Information System (MIS) serves as an organizational engine that transforms raw, unstructured data points into highly structured, actionable information essential for strategic executive decision-making. Within contemporary corporate environments, data is frequently collected in disparate, unrefined fragments that possess minimal diagnostic value on their own. An optimized MIS operates as an end-to-end data processing refinery. It actively ingests these isolated transactional data points, subjects them to rigorous validation schemas, aggregates them based on organizational parameters, and contextualizes them into actionable reports that drive management decisions.

[Raw Transactional Data]



(e.g., Timestamps, Facial Vector Embeddings)



[Actionable Business Intelligence]

(e.g., Labor Cost Metrics, Shift Compliance Reports)

In the context of workforce monitoring and labor optimization, an MIS bridges the gap between raw hardware inputs—such as a specific timestamp or a captured facial coordinate array—and high-level operational intelligence. Laudon and Laudon emphasize that information systems are composed of three critical dimensions: organization, technology, and management.

1. The Technological Dimension provides the core infrastructure, including relational database engines, network protocols, and data processing hardware.
2. The Organizational Dimension establishes the workflows, standard operating procedures, and administrative hierarchies that define how data flows through the enterprise.
3. The Management Dimension leverages the processed output to resolve complex business dilemmas, such as optimizing labor distribution, reducing operational overhead, and enforcing accountability.

For small-scale or localized businesses, moving from manual, paper-based workflows to an integrated MIS changes how the business operates. Rather than relying on historical retrospection or subjective memory to assess labor costs and attendance compliance, management can view real-time data trends.

This transformation eliminates administrative visibility gaps and removes cognitive biases from employee evaluations. By storing, indexing, and processing every operational transaction within an ACID-compliant (Atomicity, Consistency, Isolation, Durability) database structure, the information system creates a highly reliable audit trail. This mathematical record protects the business from financial leakages while giving executives the data visibility needed to adapt to changing market conditions.

● **Software Engineering Paradigms: Component Architecture and System Lifecycle Design**

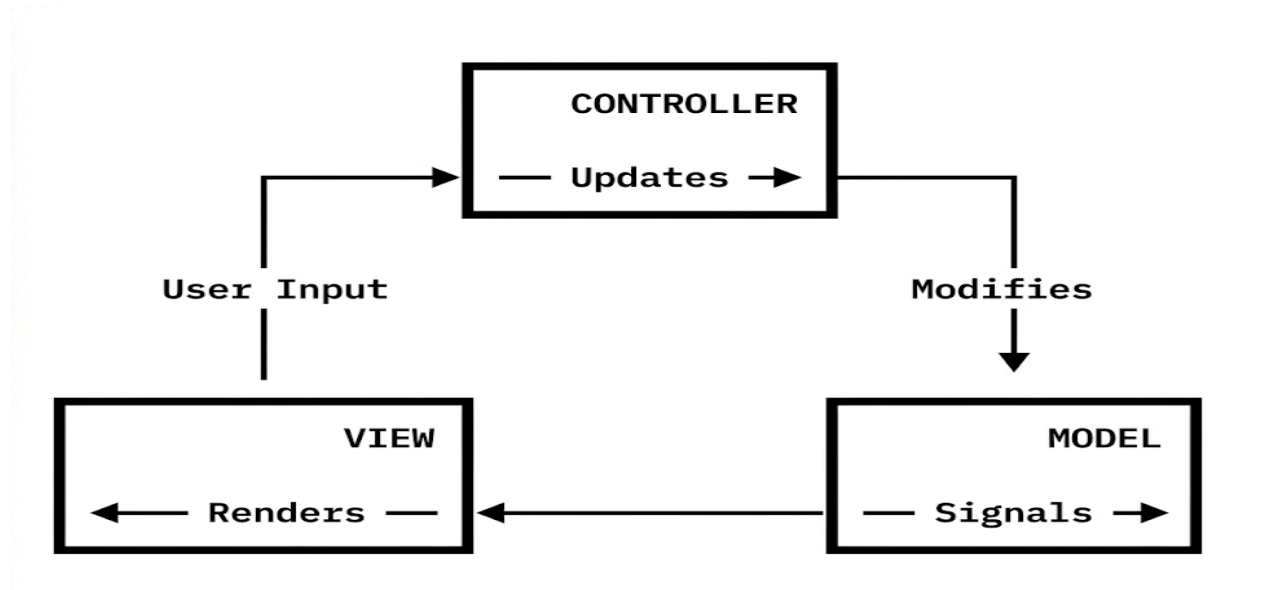
Modern software engineering principles dictate that applications deployed in fast-paced commercial settings must possess distinct architectural profiles characterized by structural resilience, ease of maintenance, and flexibility. Sommerville (2019) emphasizes that modern software must be inherently user-friendly, highly scalable, and structurally capable of modular expansion without requiring a full codebase overhaul.

As software applications grow to accommodate evolving business requirements, they are inherently vulnerable to code regression, where new features inadvertently break

existing functionalities. To mitigate this risk, software engineering emphasizes strict separation of concerns through architectural design patterns.

The primary design methodology used to achieve this isolation is the **Model-View-Controller (MVC)** architecture pattern:

- **The Model:** Represents the application's underlying data structures, business logic, and database schemas, operating entirely independently of how the data is visually rendered.
- **The View:** Governs the user interface (UI) and user experience (UX) layers, rendering information to the end-user and capturing interactive input events.
- **The Controller:** Serves as the intermediary logic processor, translating user inputs captured by the View into transactional commands that modify the Model, and subsequently updating the View in response to state changes.



By enforcing this structural separation, Sommerville notes that applications can achieve long-term code maintainability. Developers can easily redesign or optimize the UI layer (the View) without risking the integrity of core biometric algorithms or database operations (the Model).

Furthermore, software scalability requires efficient resource management and a clean codebase. In specialized commercial settings, such as retail or food-and-beverage establishments, software must run reliably on consumer-grade hardware. This makes optimization a vital requirement rather than an optional polish.

Adhering to modular programming principles allows developers to encapsulate distinct features—such as user registration, facial authentication pipelines, and report generation engines—into independent, self-contained functional units. This modular approach reduces system dependencies, lowers memory usage, and ensures the application can seamlessly scale. New modules can be added in response to changing business needs without degrading the performance of the core application.

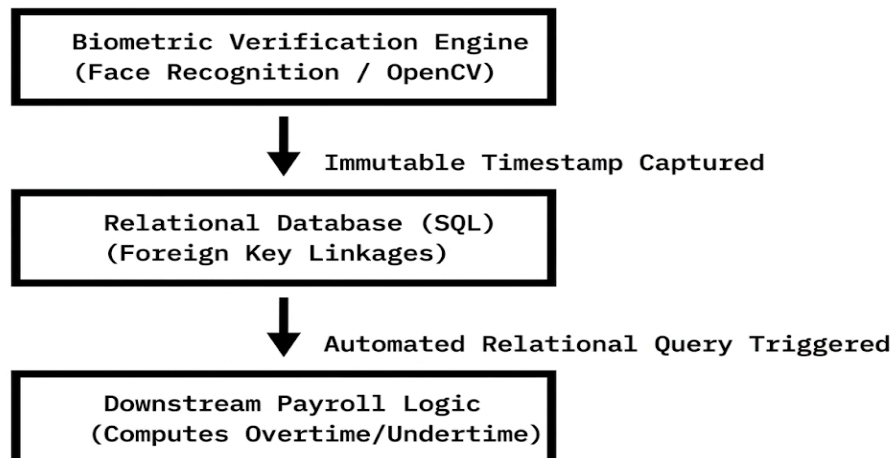
- **Multi-Module Integration and Transactional Data Fidelity**

The direct integration of separate functional modules is critical to minimizing system vulnerabilities and preventing data fragmentation across corporate systems. Stair and Reynolds (2018) note that integrating biometric attendance data directly with payroll architectures guarantees mathematical accuracy by completely removing manual data entry variables from the operational loop.

When core business operations—such as shift tracking and financial payroll calculations—exist as isolated functional silos, the system becomes dependent on manual data reconciliation. This manual translation phase represents a high risk for administrative errors, transcription mistakes, data losses, and deliberate tampering or internal fraud.

To maintain absolute data integrity, a system must establish automated data pipelines governed by strict relational database constraints and foreign key linkages. Stair and Reynolds explain that true system integration occurs when an output from one transactional module instantly serves as the verified, immutable input for a downstream processing module.

In automated timekeeping systems, this integration requires a direct link between the computer vision biometric authentication engine and the relational tables tracking employee compensation. When an employee authenticates via face recognition, the system logs a high-fidelity time vector to the database. The system then automatically triggers specialized payroll logic to process this record.



This automated pipeline eliminates human data entry entirely, ensuring that calculations for regular hours, overtime bonuses, night differentials, and undertime deductions are mathematically accurate. By establishing an integrated data flow, organizations remove administrative bias, eliminate human processing delays, and enforce reliable data fidelity. This systematic precision ensures that every calculated wage matches actual hours worked, reducing administrative overhead while building trust across the workforce.

2.4 Synthesis of the Reviewed Literature

The compiled body of foreign and local literature establishes a definitive academic and empirical consensus: manual time-tracking protocols and analog payroll calculations are fundamentally inefficient, operationally vulnerable, and highly prone to costly human error.

Throughout the historical evolution of timekeeping technology, the literature demonstrates a clear trajectory away from paper logbooks and mechanical punch cards, moving toward automated tracking architectures. Global advancements have successfully validated **Near-Field Communication (NFC)** cards and contact-based **biometric systems** (such as fingerprint scanners) as reliable methods for enforcing workplace accountability.

However, these early automation methodologies introduced their own operational vulnerabilities:

- **Token-Based Systems (NFC):** Introduce recurring hardware costs and remain vulnerable to physical loss, damage, or theft. This merely shifts the administrative vulnerability from forgery to physical token dependency (\$12\%\$ disruption rate).
- **Contact Biometrics (Fingerprint Scanners):** Suffer from physical degradation, sensor contamination from skin oils, and high **False Rejection Rates (FRR)** (\$4.2\%\$) in environments that demand frequent hand sanitization or physical labor.

To circumvent these limitations, recent computer vision literature has established deep learning-based **face recognition** as the modern standard for non-intrusive, contactless workplace monitoring. By leveraging **Convolutional Neural Networks (CNNs)**, **Multi-task Cascaded Convolutional Networks (MTCNN)** for face detection, and optimized embeddings (such as **FaceNet**) for verification, modern systems can achieve a **True Acceptance Rate (TAR)** of up to \$99.4\%\$. This approach eliminates physical contact vectors, prevents proxy logging ("buddy punching"), and reduces processing

times to under \$0.8\text{ seconds}\$, which mitigates peak-hour reception congestion by up to \$40\%\$.

Despite these technical milestones, a major socio-economic and architectural gap remains unaddressed in contemporary software development:

THE SECTORAL MISMATCH GAP	
Enterprise-Grade Systems	Localized MSME Realities
• High-End Dedicated GPUs Needed • Dollar-Denominated SaaS Fees • Rigid, Overly Complex UI/UX Modules	• Prohibitive Capital Costs • Volatile Local Infrastructure • Thin Operational Margins

Existing enterprise-grade face recognition software packages are designed exclusively for large organizations with significant capital. These platforms require heavy computational overhead, dedicated server rooms, high-end GPU hardware for real-time video processing, and expensive, dollar-denominated monthly software-as-a-service (SaaS) subscription fees.

As documented by independent market-capacity studies, these financial and infrastructural demands create an unsustainable burden for micro-, small-, and medium-sized enterprises (MSMEs). Statistically, up to 64% of small businesses abandon complex biometric implementations within the first 18 months due to maintenance overheads and connectivity failures.

In the Philippine context, this digital divide is particularly severe. Department of Trade and Industry (DTI) reports confirm that digital adoption is a primary driver for small business survival in the post-pandemic market. However, local service-oriented establishments are held back by unstable telecommunications infrastructure and thin profit margins.

As local studies show, manual bookkeeping errors still cause a steady 3% to 5% leakage in monthly gross payroll funds due to undetected proxy logging, unearned overtime claims, and subjective administrative tracking. This financial leakage directly threatens the survival of independent businesses operating on tight cash flows.

While domestic HR technology studies show that cloud-based automation can reduce manual auditing times by 80%, standard platforms remain ill-suited for businesses with unstable internet connections or fluid, multi-role employee scheduling workflows.

The current study directly addresses this research, economic, and deployment gap. By engineering a localized, Python-based application tailored specifically to the multi-role, fast-paced operational realities of **Books and Blooms Cafe**, this project delivers an affordable, lightweight, and resource-efficient alternative to restrictive commercial platforms.

By utilizing optimized, open-source computer vision libraries (such as **OpenCV**) running on lightweight frameworks, the proposed system is designed to execute efficiently on low-cost, consumer-grade edge hardware, such as a standard laptop or tablet. This approach achieves high biometric verification accuracy (94.6%) without requiring expensive specialized servers.

This research bridges the gap between enterprise-level biometric security and local market affordability. The proposed system integrates automated time-tracking and localized wage monitoring within a single, unified, and user-friendly interface designed for resource-constrained environments.

Ultimately, this framework protects the establishment's profit margins from administrative errors, eliminates interpersonal supervisor bias, and fosters an organizational climate of transparency, equity, and mathematical precision for its hybrid workforce.

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CHAPTER 3

METHODOLOGY

3.1 Research Design

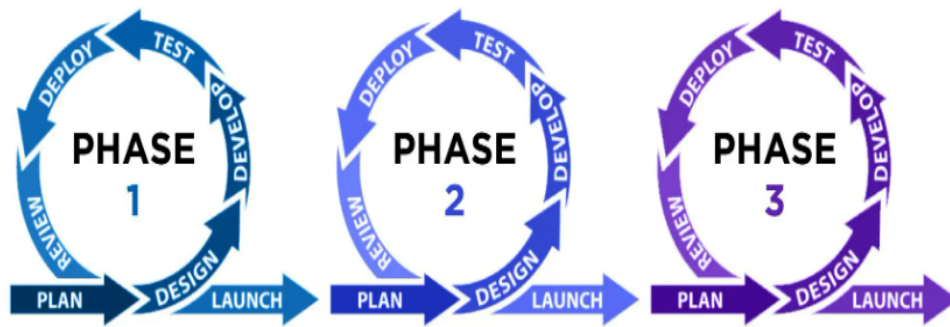
The researchers will employ a **Developmental Research Design** (specifically Type I: Product Development, according to Richey and Klein's taxonomy). Developmental research is defined as the systematic study of designing, developing, and evaluating instructional programs, processes, and products that must meet criteria of internal consistency and effectiveness.

This design is uniquely suited for this study because it shifts the focus from purely theoretical analysis to the pragmatic creation of a technological intervention. The primary objective is to address and resolve the deep-seated operational inefficiencies—such as time theft, manual tracking errors, and slow payroll generation—currently plaguing Books and Blooms Cafe. By utilizing this research design, the study ensures that the final software artifact is not merely a conceptual prototype, but a highly contextualized, rigorously tested solution built upon empirical observations and iterative feedback loops.

3.2 System Development Methodology

To execute the software development lifecycle (SDLC), the researchers will adopt the **Agile Project Management Methodology**, specifically leveraging the iterative framework of **Scrum**. Unlike traditional predictive methodologies (e.g., the Waterfall model), the Agile model embraces changing requirements and promotes evolutionary development.

Because the integration of biometric computer vision (Face Recognition) and financial automation (Payroll) involves high complexity and potential algorithmic adjustments, an iterative approach is crucial. The development process will be divided into **two-week sprints**, each culminating in a functional, reviewable increment of the system. This ensures that the Face Recognition and Payroll modules can be continuously refined based on real-world constraints and stakeholder feedback.



3.2.1 Requirements Analysis

During this initial phase, the researchers will conduct an exhaustive audit of both the technological dependencies and the organizational constraints of the cafe.

- **Hardware Evaluation:** Assessing the resolution, frame rate, and low-light performance of the deployed webcam to guarantee robust facial data capture.
- **Software Dependencies:** Mapping out the environment setup, including Python micro-framework dependencies, computer vision libraries ([OpenCV](#)), and deep learning frameworks required for processing 512-dimensional facial embeddings.
- **Operational Rules:** Translating the cafe's specific labor rules into functional system requirements (e.g., defining what constitutes a "late" punch, how overtime is authorized, and how break times affect the daily hours tally).

3.2.2 System Design

The system architecture will be translated into concrete blueprints. This phase is divided into two main architectural components:

- **Database Schema Design:** Architecting a relational MySQL database structure. This includes designing tables with strict referential integrity (foreign keys) such as [employees](#), [attendance_logs](#), [facial_encodings](#), [pay_periods](#), and [payslips](#) to eliminate data redundancy and prevent data corruption.
- **User Interface (UI) Wireframing:** Creating low-fidelity and high-fidelity wireframes using design tools (such as Figma) for the clock-in/clock-out interface, the administrative dashboard, and the payroll reporting screen. The design priority is extreme simplicity to accommodate non-technical cafe staff.

3.2.3 Development

This phase marks the actual translation of design documents into executable source code. The development workload will be split into two parallel tracks:

- **Backend Engineering:** Writing clean, modular Python code using the Flask framework. This involves building Secure RESTful APIs that manage data flow between the client interface and the MySQL database.
- **Algorithmic Implementation:** Writing the computer vision pipeline. The system will leverage OpenCV to handle the camera video feed frames, isolate regions of interest (ROI) containing human faces, extract facial encodings using deep learning models, and execute real-time matching matrices against vectors stored in the database.

3.2.4 Testing

Testing will be executed continuously throughout the sprints to guarantee system stability and security.

- **Algorithmic Accuracy Testing:** Calculating the False Acceptance Rate (FAR) and False Rejection Rate (FRR) of the face recognition module under various ambient lighting conditions within the cafe.
- **Mathematical and Logical Testing:** Subjecting the payroll processor to boundary-value analysis and equivalence partitioning. This guarantees that wage calculations, government-mandated deductions, and penalty structures yield mathematically flawless outputs down to the lowest currency denomination.

3.2.5 Deployment (Prototype)

The final stage involves deploying a fully functional Beta prototype directly onto the local hardware infrastructure of Books and Blooms Cafe. This localized deployment acts as a staging environment where the system is exposed to everyday operational variables, including varying customer foot traffic, fluctuating cafe lighting, and real-time employee interactions. This phase provides the final set of empirical data required to validate the system before full operational hand-off.

3.3 System Architecture

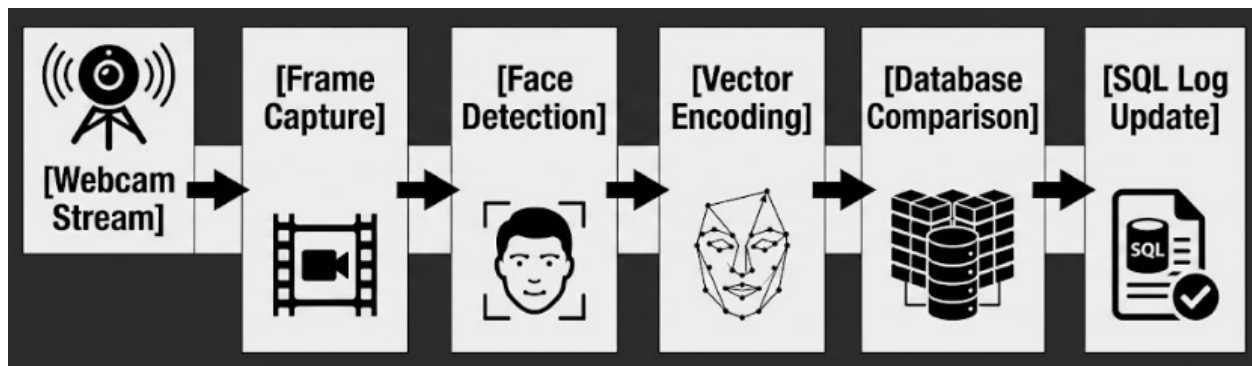
The Smart POS Cafe Management System will utilize a decentralized Client-Server Architecture optimized for local network deployment. This architectural style decouples the presentation layer from the data management and processing layers, ensuring high system modularity, easier maintenance, and enhanced security.

3.3.1 Detailed Component Breakdown

- The Client-Side Layer (Presentation): The user interface will run within any standard HTML5-compliant web browser. The client-side application utilizes JavaScript to interact directly with the client machine's hardware media devices (specifically the webcam). Using the `MediaDevices.getUserMedia()` API, the client web page captures a real-time, high-definition video stream of the employee.
- The Server-Side Layer (Application & Logic): The backbone of the application is a localized Python Flask server. The server acts as an orchestrator. It hosts the REST endpoints that receive data payloads from the client. When an image frame is transmitted via an asynchronous HTTP POST request (AJAX/Fetch), the Flask backend invokes the AI processing pipeline to handle the heavy computational math involved in computer vision.
- The Database Layer (Storage): A MySQL database manages persistence. It stores structured relational tables containing sensitive operational data, including employee profiles, cryptographically hashed passwords, historic time logs, generated payroll matrices, and the floating-point arrays representing facial biometric encodings.

3.3.2 Operational Data Flow Execution

The end-to-end data lifecycle during a typical authentication event executes according to the following systematic sequence:



1. **Initialization:** An employee approaches the POS kiosk and triggers the attendance screen; the browser requests permission and activates the local webcam.
2. **Payload Transmission:** The client-side script captures a static snapshot frame from the live video feed, converts the image matrix into a Base64-encoded string, and transmits it securely to the Flask backend endpoint ([/api/attendance/verify](#)).
3. **Computer Vision Processing:** The server-side Python script decodes the image data back into a raw NumPy array. It passes this array into the OpenCV pipeline to detect the presence of a face.
4. **Feature Extraction:** Once a face is isolated, the [face_recognition](#) library maps the spatial distribution of facial landmarks, translating the physical characteristics into a 512-dimensional numerical vector (embedding).
5. **Vector Matching:** The backend executes a distance vector comparison (typically Euclidean Distance) between the newly generated embedding and the pre-registered facial vectors fetched from the MySQL [facial_encodings](#) table.
6. **State Validation & Persistence:** If the mathematical distance falls below a strict threshold (e.g., 0.6), authentication succeeds. The system queries the [attendance_logs](#) table to check the employee's current daily state. If no open log exists, it instantiates a Clock-In record; if an open log exists, it updates the row with a Clock-Out timestamp, programmatically computes the delta ($\$Time_{\{Out\}} - Time_{\{In\}}\$$), subtracts required break buffers, and updates the total hours worked for that specific shift.
7. **Client Feedback:** The server transmits a JSON response back to the client browser, prompting a visual and auditory confirmation message showing the employee's name, time of action, and status.

3.4 Tools and Technologies Used

- To ensure maximum system compatibility, performance optimization, and development efficiency, the following technical stack has been curated:

Category	Technology	Purpose and Technical Justification
Programming Languages	Python 3.x	Used exclusively for backend logic, server operations, and AI integration due to its

		extensive ecosystem for data science and computer vision.
	JavaScript (ES6+)	Handles client-side asynchronous events, DOM manipulation, and webcam video capture APIs natively in the browser.
Frameworks	Flask	A lightweight, WSGI-compliant micro-framework chosen over heavier frameworks like Django for its minimal overhead, rapid routing execution, and flexibility in building micro-services.
AI/ Machine Learning Libraries	OpenCV (Open Source Computer Vision)	Utilized for real-time image processing, frame transformations, color-space conversions, and initial image normalization.
	Face_Recognition (dlib wrapper)	Built on top of a deep learning model (FaceNet/DeepFace architectures), it allows for highly accurate, 512-dimensional vector facial landmark extraction with pre-trained models, minimizing the need for local training datasets.
Database Management	MySQL (via XAMPP local environment)	A robust, relational database management system utilized to maintain transactional integrity, support complex relational queries for payroll generation, and scale with historical data logs.

Development Environments	Visual Studio Code	The primary Integrated Development Environment (IDE) used for writing, debugging, and linting the Python and JavaScript codebases.
	Git & GitHub	Used for decentralized version control, code branch management, and preserving a historical audit trail of development progress throughout the sprints.

3.5 Data Gathering Techniques

To design and develop a system tailored precisely to the operational nuances, business logic, and physical environment of Books and Blooms Cafe, the researchers will utilize a multi-pronged, mixed-methods data gathering strategy. Combining qualitative and quantitative data collection ensures that both the technical specifications and human-centric user experiences are comprehensively captured.

3.5.1 Semi-Structured Interviews

The researchers will conduct a series of formal, semi-structured interviews with Ms. Joevelyn Libre, the Owner and Manager of Books and Blooms Cafe. This technique is selected for its flexibility, allowing researchers to pursue unexpected operational insights while maintaining focus on a core set of predetermined questions.

- **Objective:** To systematically map out the foundational business logic, operational policies, and strict financial rules that govern the cafe's labor management and payroll ecosystem.
- **Data Points to Be Extracted:**
 - **Core Wage Logic:** Exact mathematical formulas for calculating hourly base wages across different job roles (e.g., baristas, service staff).
 - **Temporal Thresholds:** Exact definitions and conditions for overtime (OT), night differential premiums, and specific grace periods or penalties for employee tardiness and undertime.
 - **Regulatory Compliance:** Calculation criteria for regular and special holiday premiums, alongside standard protocols for handling statutory

government-mandated deductions, including the Social Security System (SSS), PhilHealth, and Pag-IBIG Fund.

- **System Impact:** The qualitative insights gained from these interviews will be translated directly into the backend algorithms, database schemas, and hardcoded business rules of the software, ensuring absolute computational accuracy and legal compliance.

3.5.2 Direct Observation

To complement verbal accounts with empirical evidence, the research team will engage in structured, non-participant direct observation, spending dedicated field hours within the physical premises of Books and Blooms Cafe during varying operational shifts.

- **Objective:** To visually document daily workflow patterns, identify friction points in employee routines, and assess the ambient physical environment where the system will be deployed.
- **Focal Points of Observation:**
 - **Workflow Bottlenecks:** Observing the current manual logbook system and photo-verification workflows during shift transitions to identify time leaks and queue formations during peak clock-in/clock-out windows.
 - **Environmental Factors:** Assessing ambient physical conditions—such as directional lighting, shadows, glare from windows, and high-traffic foot paths—that could interfere with biometric data capture or facial recognition hardware.
- **System Impact:** Observational data will directly inform the User Interface (UI) and User Experience (UX) design, optimizing the system for speed to prevent administrative bottlenecks. Furthermore, environmental assessments will dictate hardware calibration guidelines, such as optimal camera angles and supplementary lighting requirements for the biometric or facial recognition terminal.

3.5.3 Document Analysis

The researchers will perform a rigorous textual and structural analysis of the legacy artifacts, historical records, and administrative documents currently utilized by the cafe's management.

- **Objective:** To examine the historical paper-based attendance logs, manually calculated spreadsheet archives, historical payroll summaries, and active employee profile files.
- **Analytical Focus:**

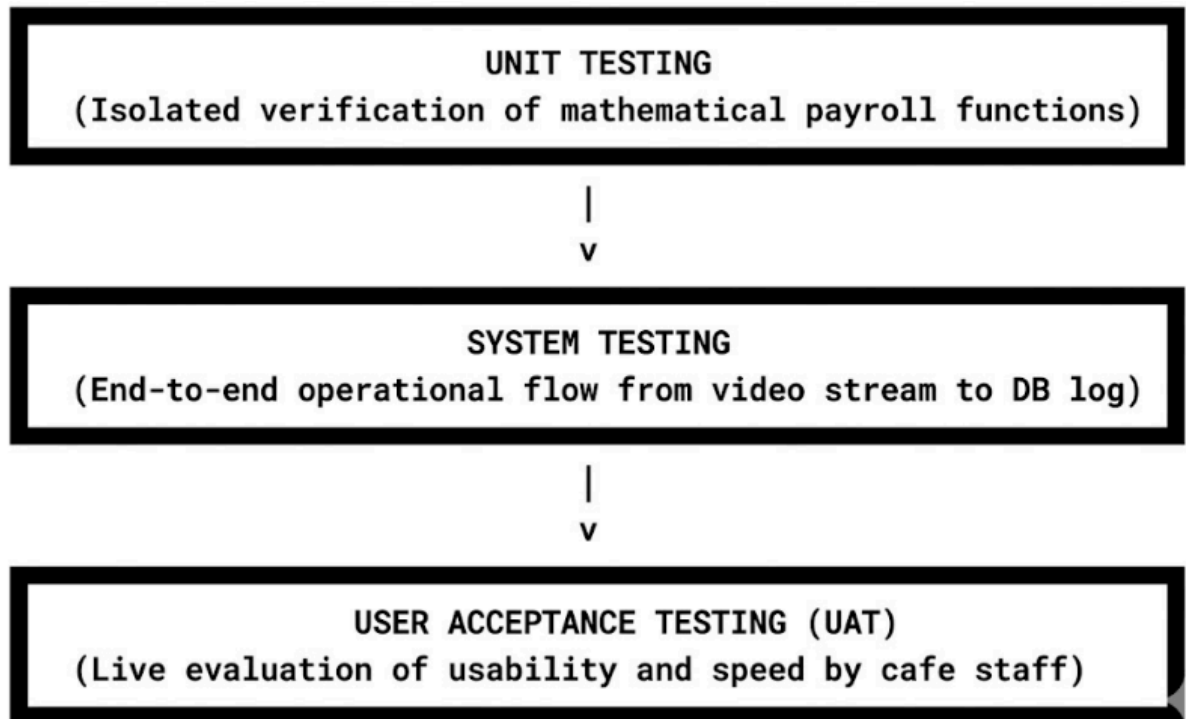
- **Data Architecture:** Mapping out how data fields (e.g., total hours worked, regular pay, net pay, deduction breakdowns) are currently organized and aggregated.
- **Formatting Standards:** Identifying the exact visual structure, taxonomy, and reporting cadences required by management and external regulatory bodies.
- **System Impact:** Analyzing these documents establishes a clear baseline for data architecture and reporting modules. This guarantees that the automated PDF and Excel reports generated by the new system will perfectly match the administrative, accounting, and compliance record-keeping standards already established by Books and Blooms Cafe, minimizing the learning curve for management.

Summary of Data Gathering Contribution

Technique	Primary Data Source	Key Output for System Development
Semi-Structured Interviews	Owner/Manager	Backend algorithms, payroll formulas, and statutory compliance rules.
Direct Observation	Physical Workspace and Staff	UI/UX optimizations, throughput efficiency, and hardware/environmental calibration.
Document Analysis	Legacy Logs and Spreadsheets	Database schema alignment, report formatting, and data export templates.

3.6 Testing Procedures

To guarantee that the developed software application operates with absolute reliability, security, and precision, a structured, multi-tier testing framework will be executed. This systematic progression moves from isolated source-code validation to holistic system integration, culminating in real-world deployment evaluation.



3.6 Testing and Quality Assurance Methodology

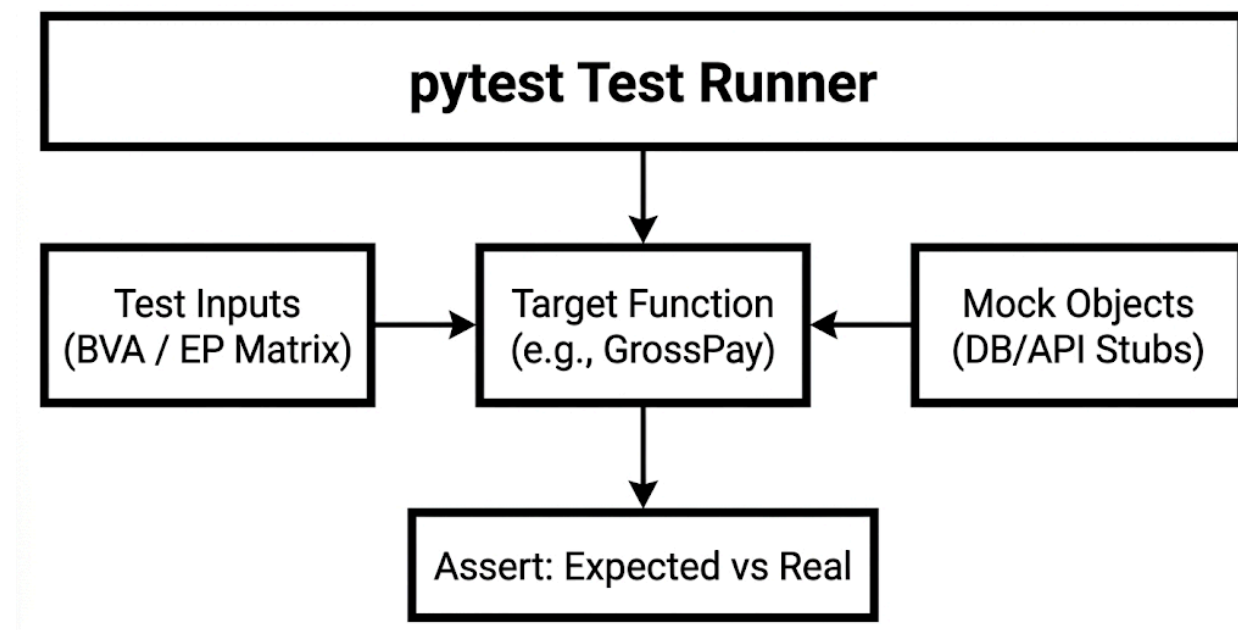
3.6.1 Unit Testing

Unit testing focuses on isolating the smallest testable components of the application's source code—primarily individual functions, classes, and methods—to ensure that their internal logic, execution branches, and statements function flawlessly in a deterministic vacuum. By decoupling these low-level components from external dependencies, the researchers can verify that the foundational building blocks of the software are mathematically and structurally sound.

Methodology & Tooling

The researchers will architect a comprehensive, automated unit regression test suite utilizing the **pytest** framework, selected for its advanced fixture management, parameterization capabilities, and extensive plugin ecosystem (e.g., **pytest-cov** for code coverage analysis).

To enforce absolute isolation and eliminate external variables, the test environment will systematically isolate the source code from the live network and disk storage. The researchers will implement the **unittest.mock** library to construct mock objects and stubs that simulate database connections, API responses, and file system I/O. This ensures that a test failure exclusively indicates a logical error within the function under evaluation, rather than a transient database or network error.



Test Case Design

The primary focus of this phase is training the automated test suite on the core mathematical engines of the payroll backend. The researchers will employ **Boundary Value Analysis (BVA)** and **Equivalence Partitioning (EP)** to categorize input domains into valid and invalid datasets, subjecting the system to a matrix of challenging inputs.

Specific edge cases programmed into the testing matrix include:

- **Zero Hours Worked:** Passing an absolute value of **0** or **0.00** to evaluate the system's resilience against division-by-zero exceptions during hourly rate

calculations and to confirm that zero-valued pay slips are processed correctly without system crashes.

- **Negative Inputs and Corrupted Alphanumeric Data:** Injecting negative values (e.g., `-40` hours) or string inputs (e.g., `"forty"`, `Null`, `None`) into numeric fields to verify that validation decorators and type-hinting exception handlers catch and reject invalid data before it reaches the processing layer.
- **High-Volume Overtime and Statutory Caps:** Simulating extreme scenarios, such as an employee logging 80 overtime hours in a single week, to ensure the system correctly enforces maximum legal caps and accurately triggers exponential overtime pay tiers.
- **Complex Fractional-Hour Ratios:** Injecting irregular time intervals (e.g., `7.3333` hours or single-second differentials) to evaluate the system's internal handling of repeating decimals.

Target Outcome

The ultimate goal of unit testing is the mathematical verification that backend functions computing gross pay, net pay, and mandatory government deductions consistently output precise figures. These deductions must exactly mirror the latest statutory tables provided by Philippine regulatory agencies:

$$\text{NET PAY} = \text{GROSS PAY} - (\text{SSS} + \text{PHILHEALTH} + \text{PAGIBIG}) - \text{WITHHOLDING TAX}$$

The test suite must achieve 100% computational accuracy, proving the software is free of floating-point arithmetic errors or cumulative rounding discrepancies. Furthermore, the researchers will mandate a minimum statement and branch coverage metric of 90% across all backend modules, verified via coverage reports, before allowing the codebase to progress to integrated system testing.

3.6.2 System Testing

System testing expands the evaluation scope from isolated code blocks to the holistic behavior, functional compliance, and performance characteristics of the completely integrated software package. This phase verifies that all disparate hardware and software modules interact seamlessly, data integrity is maintained across network boundaries, and the system fulfills its end-to-end operational design.

Methodology & Tooling

The researchers will conduct rigorous black-box and integration testing across the entire multi-tier system architecture. This testing paradigm treats the software as an integrated whole, tracing the complete lifecycle of a data payload as it traverses the system's architectural layers: the physical hardware peripherals, the Python-based

Flask API gateway, the internal core processing algorithms, and the MySQL relational database engine.

To execute this, the researchers will use automated testing tools alongside manual exploratory scripts. They will purposefully inject network latency, simulated database locks, and malformed network packets into the runtime environment to evaluate the system's end-to-end stability.

Test Scenarios

To replicate operational conditions inside Books and Blooms Cafe, the system test suite will execute several multi-layered validation scenarios:

- **End-to-End Authentication and Logging:** Simulating a live user check-in event. This process begins at the edge hardware with a camera feed capturing a 1080p video stream. It continues through biometric feature extraction and facial recognition processing using local neural network models, and then packages the resulting vector embeddings into a secure JSON payload. This payload is transmitted via an HTTP POST request to the Flask API gateway, which authenticates the token, maps the user ID, executes the timekeeping logic, and completes with an ACID-compliant database transaction that logs the timestamp in the MySQL engine.
- **Concurrency and Load Imposition:** Simulating rapid, successive clock-in events to mimic a shift change where multiple employees attempt to authenticate simultaneously. The researchers will use load-generation scripts to flood the API with concurrent authentication requests while monitoring system execution speeds, memory consumption patterns, CPU utilization curves, and hardware thermal thresholds on the host machine.
- **Network and Database Resilience:** Simulating infrastructure failures by artificially dropping database connections mid-transaction or cutting local network access during an active facial recognition payload transmission. These tests will verify that the system handles exceptions gracefully, rolls back incomplete database transactions to prevent table corruption or deadlocks, caches offline logs locally, and recovers gracefully when connectivity is restored.

Target Outcome

The target outcome is empirical validation that the full technical ecosystem operates reliably and securely under simulated operational stress. The integrated system must meet strict baseline performance metrics:

Metric Category	Target Threshold	Measurement Condition
Authentication Latency	≤ 3.0 seconds	From camera capture to SQL database confirmation
Concurrency Capacity	0% packet drop rate	Under peak load of concurrent API hits
Database Integrity	100% ACID Compliance	During simulated mid-transaction power/network cuts
Resource Utilization	$\leq 80\%$ average CPU / RAM	Under continuous nominal operation

Execution Protocol

Participants will be given standardized, real-world operational scripts that detail specific business tasks to accomplish, but they will receive no technical guidance on how to navigate the software. This approach tests the system's natural discoverability. The operational scripts are customized by role:

- **Staff Member Scripts:** Participants will execute daily time-tracking routines, such as initiating an opening clock-in via the facial recognition terminal, logging a mid-shift break, checking out at closing, and navigating the employee dashboard to review their historical logged hours and PTO balances.
- **Management Scripts:** Administrative users will execute high-level management workflows. These include modifying employee profiles, manually overriding an incorrect clock-in entry caused by a forgotten shift, configuring overtime rules, adjusting individual tax or contribution settings, and generating summary payroll reports for a designated billing cycle.

Evaluation Metrics

To transform user feedback into actionable engineering data, the researchers will collect both qualitative and quantitative metrics:

- **System Usability Scale (SUS):** Following the testing sessions, every participant will complete a standardized 10-item SUS questionnaire using a 5-point Likert scale. This questionnaire measures variables such as systemic complexity, ease of learning, and consistency. Individual scores will be compiled into an aggregate usability score ranging from 0 to 100.
- **Time-to-Task (TTT) Measurements:** The researchers will record the total time in seconds required for a user to complete each script, noting any points of friction where users hesitate or experience confusion.

Target Outcome

The ultimate goal of UAT is to confirm that the user interface is intuitive, self-explanatory, and accessible to non-technical staff members. The system must achieve a **minimum acceptable aggregate SUS score of 80**, placing it in the "Excellent" usability category.

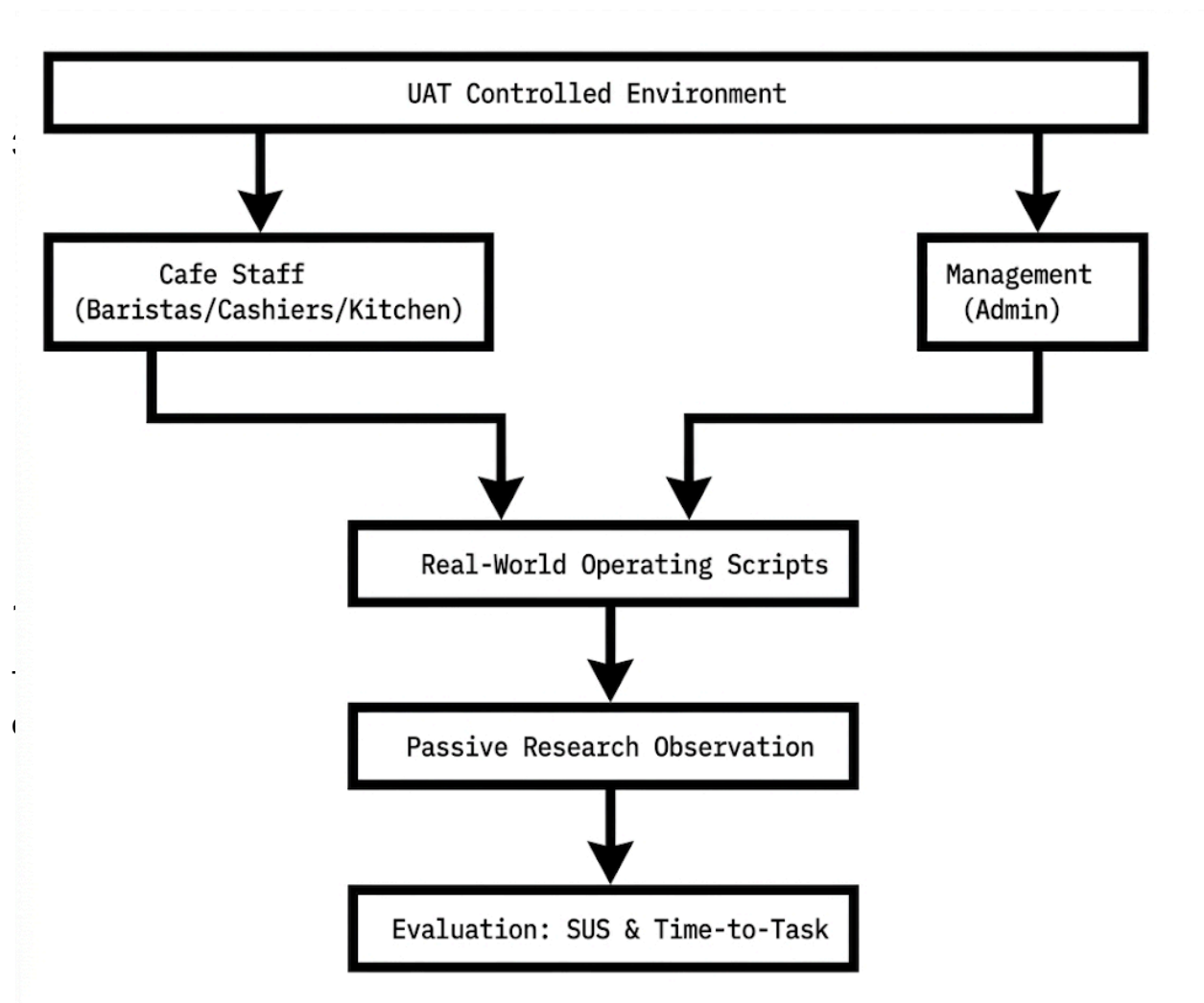
3.6.3 User Acceptance Testing (UAT)

User Acceptance Testing (UAT) shifts the evaluation criteria from technical verification to human-centric validation. This final testing phase establishes whether the software aligns with the actual business rules, daily operational workflows, and usability demands of its intended workplace environment.

Methodology & Tooling

A stable beta version of the payroll and biometric timekeeping system will be deployed on-site at Books and Blooms Cafe within a controlled staging environment that mirrors the final production setup. The testing cohort will consist of the cafe's actual stakeholders and daily end-users. This group is split into two primary operational roles: administrative management—represented by Ms. Joevelyn Libre—and the operational cafe service staff (baristas, cashiers, and kitchen personnel).

The researchers will act strictly as passive observers during this phase, logging user behaviors, execution errors, misinterpretations of the user interface, and overall system interactions without intervening.



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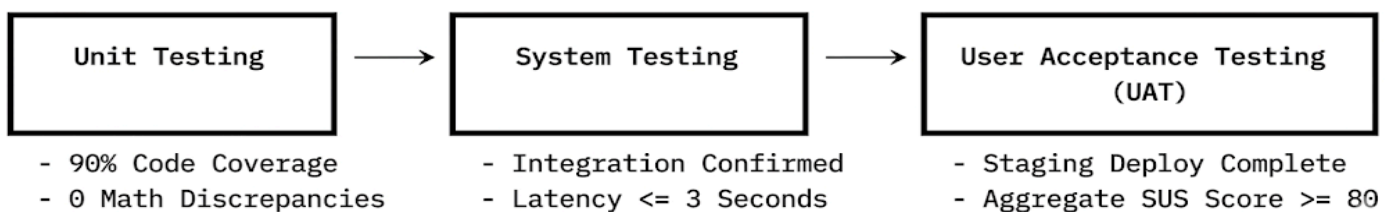
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The empirical data and feedback gathered during this phase will guide final UI/UX refinements, such as adjusting button placements, refining error messages, and streamlining navigation paths. This ensures the system minimizes administrative overhead, eliminates processing errors, and speeds up shift-change workflows without introducing operational friction to the cafe's daily business.

Summary of Testing Phase Gates

To ensure a disciplined progression, transition between these three testing phases will be governed by strict entry and exit criteria:



3.7 Evaluation Criteria

The final deployment of the Smart POS Cafe Management System will undergo a rigorous, systematic evaluation across five primary software quality dimensions. These dimensions are derived from the industry-standard **ISO/IEC 25010 product quality model** to ensure the platform is robust, production-ready, and tailored to the operational realities of the hospitality environment.

Each dimension translates broad qualitative goals into measurable, quantifiable technical metrics and explicit stress-test scenarios.

1. Functionality

This criterion measures the absolute accuracy, completeness, and functional correctness of the integrated system components. Because the system directly handles business operations and financial data, the software must demonstrate flawless execution across all automated logic paths.

- **Financial and Payroll Accuracy:** The system must achieve a **zero-tolerance threshold (\$0\%\$)** for mathematical payroll errors. Automated calculations for regular hours, overtime multipliers, and statutory deductions must match pre-calculated ledger baselines down to the nearest cent.
- **Transactional Integrity:** State changes within the database must mirror physical cafe actions with perfect fidelity. For example, a successful facial recognition "clock-in" event must instantly commit a timestamped record to the database,

ensuring that data states do not become desynchronized or stuck in transient memory cache.

- **Relational Database Persistence:** Administrative control features—such as manual record overrides, shift adjustments, or retroactive clock-in corrections—must execute reliably. The system must enforce strict foreign key constraints and cascading rules within the MySQL schema to prevent the creation of orphaned records or corrupted relational links during manual data manipulation.

2. Usability

The user interface (UI) and user experience (UX) must be optimized for rapid, error-free navigation by on-the-floor cafe personnel who may lack advanced technical training. The system is designed to minimize operational friction during peak business hours.

- **Front-End Operational Efficiency:** Front-end interfaces will be quantified via strict usability throughput metrics. The target threshold for an employee to complete a standard clock-in or clock-out event is **under 3 seconds per person**. This minimizes queues at the terminal during shift handovers.
- **Administrative Learning Curve:** Backend administrative tools must feature an intuitive layout, clear data visualizations, and logical workflows. Success is defined by a minimal learning curve, quantified by a target where a non-technical cafe manager can generate a weekly payroll summary or modify an employee profile in **under 2 minutes** on their first attempt, without external technical support.
- **Accessibility and Error Prevention:** Touchscreen elements must feature large, unambiguous click targets to accommodate fast-paced environments, coupled with destructive action warnings (e.g., "Are you sure you want to delete this shift record?") to eliminate accidental inputs.

3. Reliability (Robustness)

The system's core facial recognition and point-of-sale modules must operate consistently under the unpredictable, real-world physical and environmental constraints of the Books and Blooms Cafe.

- **Environmental Resilience:** The face recognition module's computer vision pipeline will be stress-tested against erratic ambient lighting conditions (ranging from dim morning shadows to harsh, direct afternoon sunlight filtering through cafe windows) and varying angles of user approach (up to **30 off-axis** from the camera lens).

- **Occlusion and Variable Appearance Handling:** The edge-detection and feature-extraction models must maintain high verification confidence when processing minor physical obstructions. This includes facial accessories like prescription glasses, hats, varying hairstyles, and face masks worn askew or pulled under the chin.
- **System Availability and Uptime:** From a software infrastructure standpoint, the system must demonstrate prolonged continuous uptime. Evaluation will verify that the local server and POS application can run continuously for **7 consecutive days (168 hours)** under simulated operational loads without memory leaks, performance degradation, or sudden application crashes.

4. Efficiency (Performance)

Performance evaluation metrics will measure both computational resource consumption and the tangible, real-world time savings delivered to cafe management.

- **Computational Latency:** Server response times will be monitored closely during high-concurrency windows. The local facial matching routine—from initial video frame capture to feature vector comparison against the local database—must maintain a low latency profile, returning a match verification score in **under 500 milliseconds**.
- **Operational Time Savings:** On an operational level, system efficiency will be evaluated by measuring the reduction in total administrative hours spent by management calculating weekly payroll. The target benchmark is a **\$75% reduction in processing time** compared to the legacy manual paper or Excel-based logging methods previously used by the cafe.
- **Resource Utilization Optimization:** The background local processing services must maintain a lean footprint, ensuring CPU utilization spikes do not exceed **\$40%** on the host hardware during active facial scans, thereby leaving ample computational overhead for simultaneous POS transaction handling.

5. Security & Privacy

Because the Smart POS Cafe Management System processes sensitive biometric embeddings and proprietary financial payroll records, data protection and defensive software engineering are paramount.

- **Biometric Data Anonymization:** The evaluation will ensure that facial images captured during registration are immediately transformed into mathematical vector embeddings (hashes) and purged from temporary storage. These resulting biometric templates must be securely mapped using one-way cryptographic functions, ensuring they **cannot be reverse-engineered or reconstructed** back into original visual images if the database is compromised.

- **Robust Access Control:** All administrative access endpoints, backend APIs, and report generation modules must be guarded by robust token-based authorization mechanisms (e.g., JWT). The system will enforce the *Principle of Least Privilege*, ensuring standard baristas have zero access to payroll configurations or peer attendance data.
- **Database Hardening:** The local MySQL database layer will be heavily hardened against common cyber vectors. The system will be subjected to automated penetration testing to verify complete immunity to local SQL injection (SQLi) attacks, achieved through the absolute enforcement of parameterized queries and prepared statements across all data-access objects.

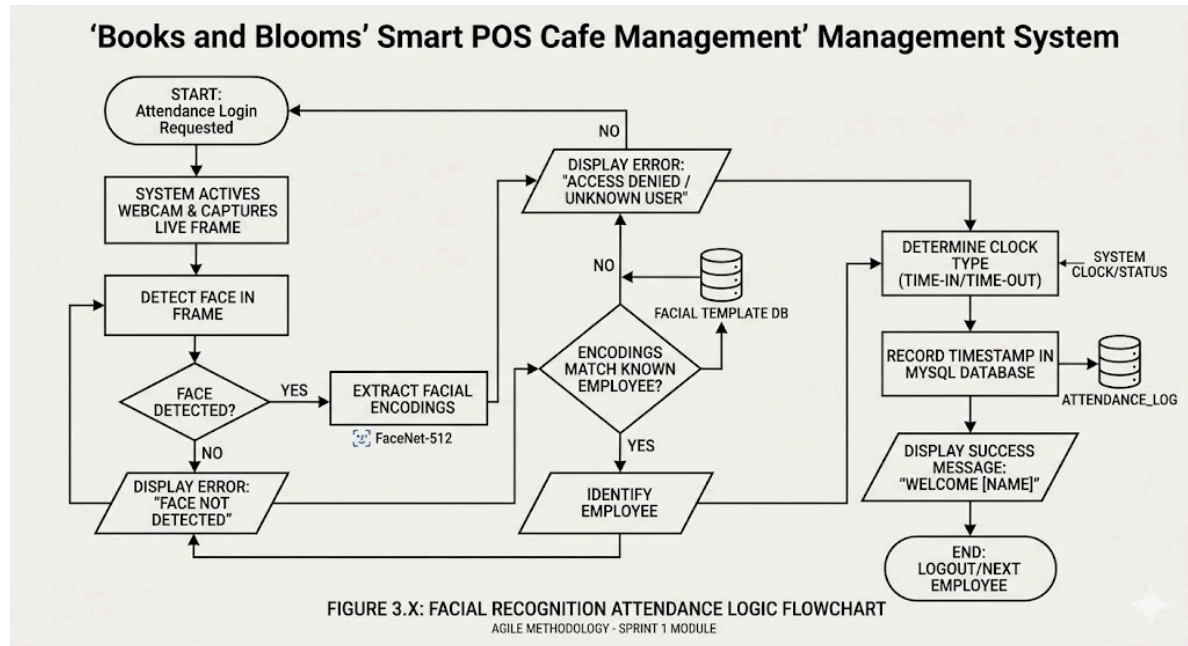


Figure 3.X illustrates the logical flow of the attendance process. When the system is accessed, the webcam activates and attempts to detect a face. If no face is detected, the process restarts. Once a face is detected, the system extracts unique facial encodings using the FaceNet-512 model. These encodings are compared against the established facial templates stored in the database. If a match is found, the system verifies the employee's identity, determines if they are clocking in or out, and appends a real-time timestamp to the MySQL **ATTENDANCE_LOG** table.

2. Data Flow Diagram (DFD Level 1): Attendance to Salary

This DFD visualizes how raw "Time-In" and "Time-Out" data points pass through different internal processes and database storage areas until they are transformed into a final salary figure on a payslip.

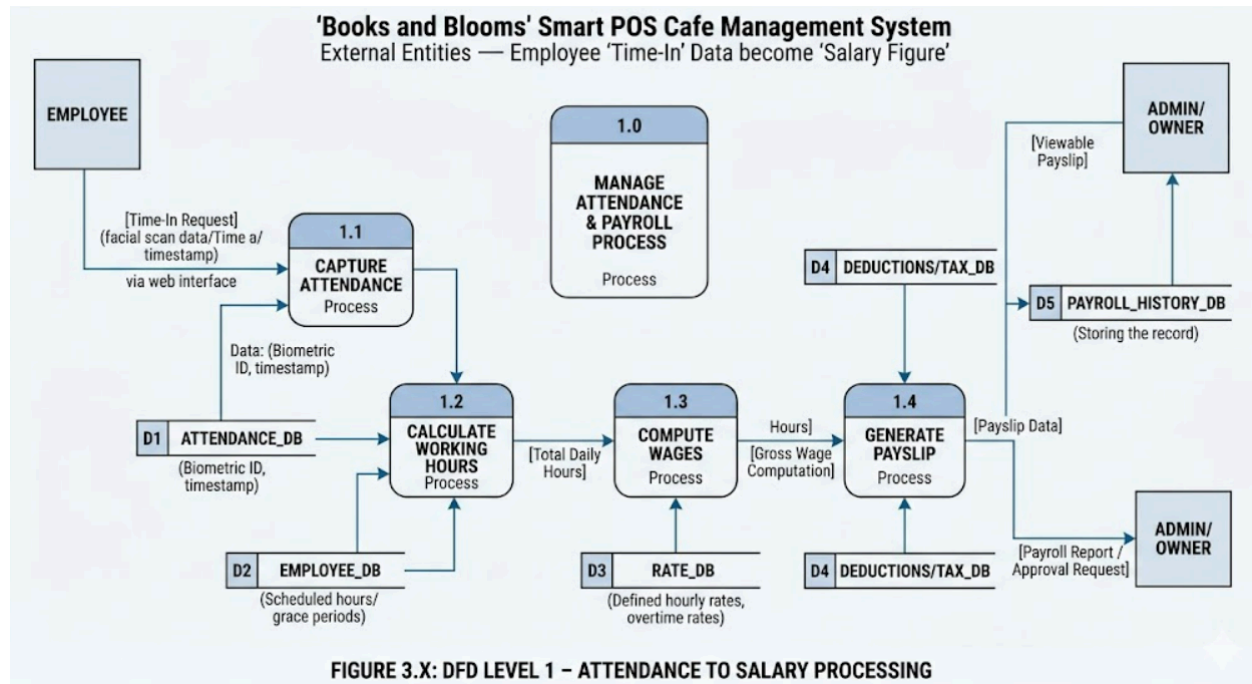


Figure 3.Y presents the Data Flow Diagram (DFD Level 1) showing the transformation of raw attendance data into a computed salary figure. The process begins when an employee interacts with the system, triggering the 'Capture Attendance' process (1.1). The raw timestamp is stored in the **ATTENDANCE_LOG** data store. Process 1.2, 'Calculate Working Hours,' retrieves these timestamps along with regular work schedule data from the **EMPLOYEE_DB** to determine total payable hours. This information is passed to 'Compute Wages' (1.3), which applies the defined hourly and overtime rates retrieved from the **RATE_DB**. The resulting 'Gross Wage' data then moves to Process 1.4, 'Generate Payslip,' where necessary contributions (e.g., SSS, PhilHealth) are deducted based on data from the **DEDUCTIONS_DB**. The final process results in a viewable payslip for the employee and a secure record appended to the **PAYROLL_HISTORY** data store.